SHAREPOINT 2016 → POWER AUTOMATE

Moving SharePoint 2016 workflows to Power Automate

High-level plan and conversion pipeline

Version 1.0 · October 2026

How we will move our SharePoint Server 2016 workflows (Nintex Workflow 2016 and SharePoint Designer) to Microsoft Power Automate: why, in which order, with which method, and who does what.

Why

SharePoint Server 2016 and InfoPath 2013 have been out of support since 14 July 2026. SharePoint Online no longer runs the old workflows, so they must be rebuilt.

How

Every workflow goes through the same nine-stage pipeline: rebuild it with the same logic, prove it with side-by-side tests, then improve it.

Where we are

The pilot, *Proposal Approval*, is analysed and its rebuild guide is ready. Next: platform decisions, missing Nintex exports, inventory.

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1. Summary

- **Why.** SharePoint Server 2016 and InfoPath 2013 have been out of Microsoft support since 14 July 2026. SharePoint Online no longer runs the older SharePoint 2013 workflows (since 2 April 2026), so our workflows cannot simply be moved to the cloud as they are. Our Nintex workflows depend on SharePoint 2016, so they have to be rebuilt.
- **Where to.** Power Automate cloud flows, with Approvals, Microsoft Teams and Outlook for the people steps, and Power Apps where we need forms (the InfoPath forms are being replaced the same way).
- **How.** Every workflow goes through the same nine-stage **conversion pipeline** (section 4). Stages 0 to 2 happen once for the whole estate. Stages 3 to 8 are repeated for each workflow, in waves.
- **Rebuild first, improve second.** We first rebuild each workflow with **exactly the same logic**, including its known defects, and prove it with side-by-side tests. Only then do we fix the defects and improve the process. This keeps every change visible and every result testable.
- **Pilot.** *Proposal Approval*, one of our largest Nintex workflows (195 actions), is the pilot. Its analysis and a complete step-by-step rebuild guide are already done (section 11).
- **Decisions needed now:**
  1. the target for the SharePoint data (SharePoint Online, or SharePoint 2016 behind a gateway);
  2. the Power Platform environments and licences;
  3. a service account and a shared mailbox for the flows;
  4. a business owner for each workflow;
  5. getting the missing Nintex exports (user-defined actions and child workflows).

2. Why we have to move

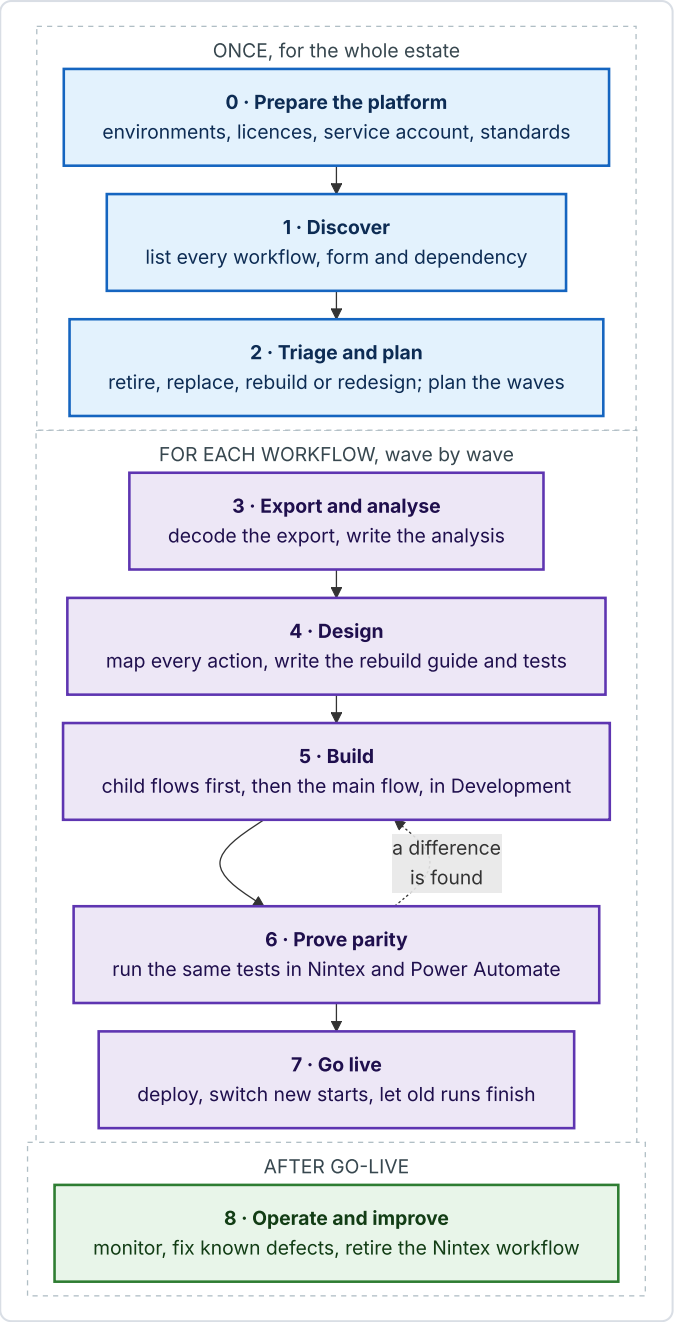
Date	What happened	What it means for us
2 April 2026	SharePoint 2013 workflows stopped running in SharePoint Online	Workflows cannot be moved to the cloud as they are. They must be rebuilt, in Power Automate
14 July 2026	Extended support ended for SharePoint Server 2016 and 2019	No more security fixes for the platform our workflows run on
14 July 2026	Support ended for the InfoPath 2013 client	The forms our workflows start from (for example the Project Establishment Form) must be replaced, for example with Power Apps
No date announced	Nintex Workflow for SharePoint 2016	Nintex has not announced an end date, but the product runs only on SharePoint 2016, which is no longer supported

Running unsupported software is a growing security and operational risk. The sooner the workflows are moved, the sooner SharePoint 2016 can be switched off.

3. Principles

1. **Same logic first.** Each workflow is rebuilt to behave exactly as it does today. Known defects are copied on purpose and marked, so nothing changes without a decision.
2. **Prove it.** A workflow goes live only when the same test scenarios give the same results in Nintex and in Power Automate.
3. **Improve after go-live.** Defects and improvements are handled as normal, tested changes once the rebuild is live.
4. **One pipeline for every workflow.** The same stages, templates and checklists every time, so the work is predictable and can be shared between people.
5. **Everything in solutions.** Flows, environment variables and connections live in Power Platform solutions, built in Development and moved to Test and Production as managed solutions.
6. **Document as we go.** Every analysis, rebuild guide and test result is kept in this repository, so knowledge does not depend on one person.

4. The conversion pipeline



Stages 0–2 run once for the whole estate. Stages 3–7 run for each workflow, wave by wave. If the parity test finds a difference, the workflow goes back to Build. Stage 8 runs after go-live.

Stage	Purpose	Main output	Done when
0 · Prepare the platform	Get the target ready	Environments, licences, service account, standards	A test flow can read and write the target lists
1 · Discover	Know what we have	Inventory of every workflow, form and dependency	Every workflow has an owner, usage figures and a size
2 · Triage and plan	Decide what happens to each workflow	Retire / replace / rebuild / redesign decision; wave plan	The sponsor has approved the wave plan

Stage	Purpose	Main output	Done when
3 · Export and analyse	Understand one workflow completely	Decoded export and analysis document	The process owner agrees the analysis is right
4 · Design	Decide how to build it	Rebuild guide with an action map and test plan	Every Nintex action has a Power Automate step
5 · Build	Build it	Child flows and main flow in a solution in Development	Every step in the guide is built and reviewed
6 · Prove parity	Show it behaves the same	Signed-off parity test results	All test scenarios match, apart from documented deviations
7 · Go live	Switch over safely	Managed solution in Production; Nintex start disabled	New requests run in Power Automate; old ones finish in Nintex
8 · Operate and improve	Keep it healthy, then make it better	Monitoring, defect fixes, retired Nintex workflow	The Nintex workflow is removed and its history archived

5. The stages in detail

Stage 0 · Prepare the platform (once)

Goal. A ready, governed place to build and run the flows.

What we do:

- **Decide where the SharePoint data will live.** The recommended target is **SharePoint Online**. If some lists must stay on SharePoint 2016 for a while, Power Automate can reach them through an **on-premises data gateway**, which needs Premium licences.
- **Create three Power Platform environments:** Development, Test and Production. Set up a deployment pipeline between them.
- **Agree licences.** Standard Microsoft 365 licences cover the SharePoint, Outlook, Teams and Approvals connectors. The gateway or premium connectors need Power Automate Premium.
- **Create a service account** that owns the connections, and a **shared mailbox** that sends the e-mails.
- **Set up governance:** a data-loss-prevention (DLP) policy, naming rules, and who may create flows in which environment.
- **Agree the building standards:** the patterns in the rebuild guide (refresh item, set field, query, log, approval request) become our team standard.

Output. Environments, licences, accounts and standards are ready.

Done when a test flow in Production can read and write a target list as the service account.

Stage 1 · Discover (once)

Goal. A complete list of what we have to move.

What we do:

- **List every workflow** on every SharePoint 2016 site: Nintex workflows, SharePoint Designer 2010 and 2013 workflows, and out-of-the-box workflows. Microsoft's *SharePoint Migration Assessment Tool* (SMAT) and Nintex's own reports can help.
- **For each workflow, record:**
  - the site and list it runs on;
  - how it starts (by hand, on create, on change);
  - how often it ran in the last 12 months;
  - the business owner.
- **List the dependencies:** user-defined actions (UDAs), child workflows, task forms, InfoPath forms, other lists it reads, web services and hard-coded addresses.
- **Run the decoder on each Nintex export** to get these facts automatically: `python3 tools/nwf-decode/nwf_decode.py --summary <workflow>.nwf`.

Output. A workflow inventory (one row per workflow) with usage, owner, dependencies and size.

Done when every workflow in the inventory has an owner, usage figures and a size.

Stage 2 · Triage and plan (once)

Goal. A decision for every workflow and a realistic order of work.

What we do:

- **Decide each workflow's fate** with the triage tree: **retire**, **replace** with a standard feature, **rebuild with parity**, or **redesign**.
- **Size it:** small, medium or large (section 6).
- **Group the workflows into waves** of similar size and shared dependencies. Workflows that share a user-defined action or a list go in the same wave, so the shared parts are built once.
- **Order the waves** by business importance and risk. Workflows on sites with the most sensitive data, or that stop the business when they fail, go first.

Output. The decision list and the wave plan.

Done when the sponsor approves the wave plan.

Stage 3 · Export and analyse (each workflow)

Goal. Understand one workflow completely before changing anything.

What we do:

1. **Export everything** from Nintex: the workflow (`.nwf`), every user-defined action it calls (*Site Settings → Manage User Defined Actions → Export*), every child workflow it starts, its forms, and the list schemas.
2. **Decode the export** with `tools/nwf-decode/nwf_decode.py`. It numbers every action (A001, A002, …), so that the analysis, the design, the build and the tests all refer to the same IDs.
3. **Write the analysis document:**
   - what the workflow does;
   - a flow diagram;
   - a decision table (who approves in which case);
   - what it writes to the item;
   - the e-mails it sends;
   - a list of defects and risks.
4. **Review it with the process owner.**

Example. docs/proposal-approval-nwf.md analyses all 195 actions of *Proposal Approval* and lists 46 issues.

Done when the process owner agrees that the analysis describes the process as it works today.

Stage 4 · Design (each workflow)

Goal. A build plan that someone else could follow.

What we do:

- **Map every Nintex action** to a Power Automate step, using the cheat sheet and the standard patterns.
- **Decide the people steps:**
  - approvals become **Approvals**;
  - data-entry tasks become **Teams adaptive cards** or a **Power App**;
  - InfoPath forms become **Power Apps**.
- **Decide what becomes a child flow.** Lookups become short child flows. Long human steps stay inline, because a child flow must answer its caller within 120 seconds.
- **List the deviations**, meaning where Power Automate cannot behave identically. Examples: the 30-day run limit, a different e-mail sender, Teams instead of task forms.
- **Mark every copied defect** with a parity note.
- **Write the parity test plan:** scenarios with the expected approvers, status texts, column values and e-mails.

Example. powerautomate/proposal-approval-flow/README.md has a step for every one of the 195 actions, 21 deviations and 26 test scenarios.

Done when every Nintex action has a step in the action map, and the test plan is agreed.

Stage 5 · Build (each workflow)

Goal. A working flow in the Development environment.

What we do:

- **Build inside a solution:** first the environment variables and connection references, then the child flows, then the main flow, stage by stage.
- **Name every action after its Nintex ID** (for example *A055 Supervisor's Approval*), so it can be traced back to the analysis.
- **Have a second builder review** the flow against the action map.

Done when every step in the guide is built and the review is complete.

Stage 6 · Prove parity (each workflow)

Goal. Evidence that the new flow behaves like the old workflow.

What we do:

- **Set up a test site** with copies of the lists. Point the Test environment at it.
- **Run every scenario** from the test plan in Nintex and in Power Automate, with the same test users giving the same answers.
- **Compare the results:** who was asked to approve, the status texts, the column values, the approval status and the e-mails.
- **Send any unexplained difference back to Build.** Differences that cannot be avoided must already be listed as deviations.
- **Get sign-off** from the process owner.

Done when all scenarios match, apart from documented deviations, and the process owner signs off.

Stage 7 · Go live (each workflow)

Goal. A safe switch with nothing lost.

What we do:

- **Deploy** the managed solution to Production. Set the environment variables, connections and run-only users.
- **Tell users** what changes for them: where to start the flow, how approvals now arrive, and the new e-mail sender.
- **Switch:** new requests start in Power Automate, and the Nintex start option is removed.
- **Let running Nintex instances finish.** Do not cancel them.
  - If the lists move to SharePoint Online at the same time, agree a short freeze. During the freeze no new requests start, the open ones finish, and then the data moves.
  - Requests still open at the end of the freeze are restarted in Power Automate.
- **Watch the first runs closely.**

Done when new requests run in Power Automate and no request is left without an owner.

Stage 8 · Operate and improve (each workflow)

Goal. A healthy flow, then a better one.

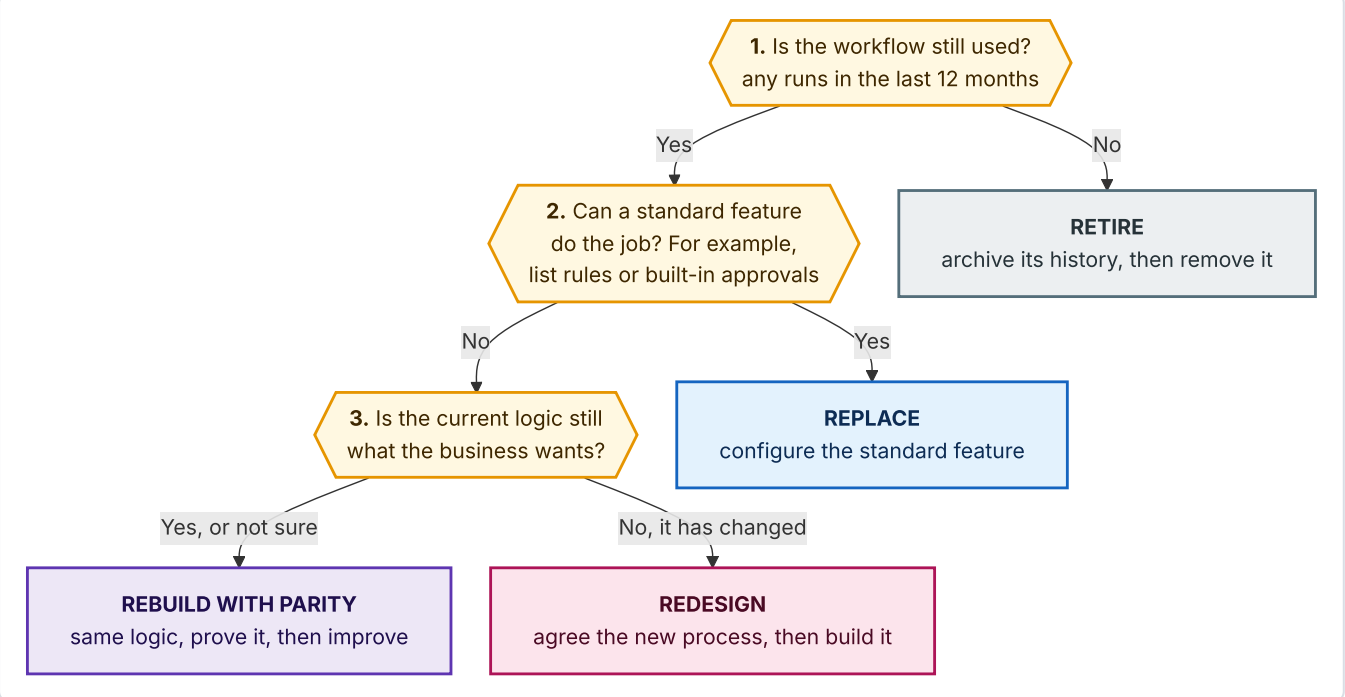
What we do:

- **Monitor** failed runs and runs older than about 20 days (a Power Automate run stops after 30 days). Agree who responds.
- **Fix the defects copied in Stage 4,** one change at a time, each with its own test. For *Proposal Approval* the first two are:
  - managers skipping higher approval (issue I01);
  - a GM approval overriding a rejection (issue I02).
- **When all open Nintex instances have finished:**
  - export and archive the Nintex workflow history;
  - remove the Nintex workflow;
  - record the workflow as done in the inventory.

Done when the Nintex workflow is removed and its history is archived.

6. Triage and sizing

Deciding what happens to each workflow



Answer the three questions in order. "Not sure" at question 3 means rebuild with parity: it is safer to copy the current behaviour and improve it later than to change it without evidence.

Decision	When	What we do
Retire	No runs in the last 12 months, or the process no longer exists	Archive the history, tell the owner, remove the workflow

Decision	When	What we do
Replace	A standard feature does the job, for example a list rule, an alert, or the built-in approval on a document library	Configure the feature; no flow to build
Rebuild with parity	The workflow is used and the logic is still right, or nobody is sure	Run stages 3–8 as described
Redesign	The business process has changed	Agree the new process first, then build it; test against the new process instead of the old workflow

Sizing

The decoder's `--summary` output suggests a size. Adjust it with the owner's knowledge of the process.

Size	Typical workflow	Planning effort, stages 3–7
Small	Up to about 40 actions; no tasks; no user-defined actions or child workflows. For example, a notification workflow	2–5 days
Medium	40–120 actions; or approval or data-entry tasks; or one child workflow	2–3 weeks
Large	More than 120 actions; or user-defined actions; or several child workflows and task forms. *Proposal Approval* is Large: 195 actions, 3 user-defined actions, 3 child workflows and 2 task forms	4–8 weeks

These efforts are **planning assumptions**. We replace them with real figures after the pilot. The elapsed time is usually longer than the effort, because testing depends on business users' time.

7. Phases and timeline

The timeline is in weeks from the start of the programme. Phases overlap on purpose.

Phase	Weeks	What happens	Result
1. Foundation and discovery	1–4	Stages 0, 1 and 2	Platform ready; inventory; approved wave plan
2. Pilot: Proposal Approval	3–12	Stages 3–8 for the pilot; stages 3 and 4 are already done	First flow live; measured effort; templates and checklists confirmed
3. Wave 1: Project Establishment Form	10–20	Its Nintex approval workflow and the InfoPath form (a Power Apps demo already exists), plus other high-priority workflows	The PEF process runs on Power Platform
4. Further waves	from week 18, every 6–8 weeks	3–6 workflows per wave, depending on size	Workflows move wave by wave
5. Decommission	after the last wave, about 4 weeks	Archive histories; remove Nintex; set SharePoint 2016 to read-only, then switch it off	SharePoint 2016 retired

The length of phase 4 depends on the size of the inventory from stage 1. We will set real dates once the inventory and the pilot's effort are known.

8. Roles and responsibilities

Role	Responsibilities
Sponsor	Sets priorities; approves the wave plan and every *retire* or *redesign* decision; removes blockers
Process owner (one per workflow, from the business)	Explains the process; confirms the analysis; supplies test users; signs off the parity test; decides on improvements
Power Platform administrator	Environments, licences, DLP policy, service account, connections, deployment pipeline, monitoring
SharePoint administrator	Inventory; Nintex exports (workflows, user-defined actions, child workflows); list migration; permissions
Flow developer	Stages 3–5: analysis, design, build; fixes found in testing
Reviewer (a second developer)	Checks each build against its action map
Testers (business users)	Run the parity scenarios in both systems
Support	First response to failed runs after go-live

9. How Nintex building blocks map to Power Automate

Nintex	Power Automate
Workflow started by hand from the item menu	Instant flow with the *For a selected file* or *For a selected item* trigger
Workflow started when an item is created or changed	*When an item is created* or *When an item or file is modified* trigger
Workflow variables	*Initialize variable* actions at the top of the flow
Action set	Scope
Set a condition / Run if	Condition
Query list	SharePoint *Get items* with a filter
Set field / Update item	SharePoint *Update item*, or an HTTP request that updates only the named columns
Build string, Set variable	*Compose*, *Set variable*
Send notification	Outlook *Send an email (V2)*, from a shared mailbox
Request approval / user-defined approval action	Approvals *Start and wait for an approval*
Request data (task form)	Teams *Post adaptive card and wait for a response*, or a Power App
Start workflow (child workflow)	*Run a Child Flow*; long-running children answer at once and report completion later
User-defined action (UDA)	A child flow, or a copied scope when it waits for people
Set approval status	SharePoint *Set content approval status*
Wait for check-out status	*Do until* loop with a delay
Log in history list	Run history, plus an optional history list
Commit pending changes	Not needed: Power Automate writes immediately
End workflow	*Terminate*
InfoPath form	Power Apps canvas app, or a customised list form

The rebuild guide shows every one of these in detail, with exact expressions.

10. Main risks

Risk	Impact	How we handle it
A Power Automate run lasts at most **30 days**; Nintex had no limit	Very slow approvals stop part-way	Reminders while waiting; monitor runs older than 20 days; restart if needed
A **child flow must answer within 120 seconds**	Long child processes time out	Use child flows only for quick steps; long ones answer at once and report completion later
Task forms are replaced by Approvals, Teams cards or Power Apps	Users work in a different place	Show users early; short guides; Teams access for every approver
Hidden dependencies: user-defined actions, child workflows, forms, hard-coded addresses	A rebuilt flow misses a step	Decoder summary in stage 1; full exports in stage 3; environment variables instead of hard-coded addresses
Data and identity differences after moving to SharePoint Online (people fields, display names, date and time zones)	Lookups or comparisons behave differently	Covered by the parity tests; listed as deviations
Running instances at the moment of switching	Requests lost or duplicated	Let them finish in Nintex; freeze before data moves; restart the few that remain
Licences and gateway if data stays on SharePoint 2016	Extra cost; Premium needed	Decide the data target in stage 0; prefer SharePoint Online
Unsupported platform while we migrate	Security exposure grows over time	Prioritise exposed and critical sites; keep the programme short; retire SharePoint 2016 at the end
Known defects are copied by design	The old mistakes continue for a short time	Parity notes make every one visible; fix them in stage 8 with their own tests
Knowledge sits with one person	Delays if that person is away	Everything documented in this repository; reviewer role in stage 5

11. Where we are now

Item	Status	Where
Proposal Approval workflow analysed (stage 3)	Done	docs/proposal-approval-nwf.md
Proposal Approval rebuild guide, diagrams, deviations and test plan (stage 4)	Done	powerautomate/proposal-approval-flow/
Decoder for Nintex exports (stages 1 and 3)	Done	tools/nwf-decode/
Project Establishment Form (InfoPath) analysed	Done	docs/template-xsn.md
Project Establishment Form as a Power Apps canvas app	Demo with sample data	powerapps/pef-canvas-app/
Exports of the 3 user-defined actions and 3 child workflows used by *Proposal Approval*	To do	SharePoint administrator
Export of the Project Establishment Form's own Nintex approval workflow	To do	SharePoint administrator
Platform decisions (data target, environments, licences, service account)	To do	Sponsor and Power Platform administrator
Inventory of all other workflows (stage 1)	To do	SharePoint administrator

The next five steps:

1. **Decide the data target** (SharePoint Online or the gateway), and create the three environments.
2. **Export from Nintex** the user-defined actions 1000009, 1000010 and 1000022, the child workflows *Check Signatures*, *PROPOSAL CHECKER* and *Generate PDF*, and the Project Establishment Form approval workflow.
3. **Run the inventory** (stage 1) across all SharePoint 2016 sites.
4. **Build the *Proposal Approval* pilot** from the rebuild guide (stage 5).
5. **Run its parity tests** (stage 6) with the process owner.

12. Checklist for each workflow

Stage 3 · Export and analyse

- ☐ The workflow, its user-defined actions, child workflows, forms and list schemas are exported
- ☐ The decoder output is saved with the export
- ☐ The analysis document is written and reviewed with the process owner

Stage 4 · Design

- ☐ Every Nintex action is in the action map
- ☐ Every copied defect has a parity note
- ☐ The deviations are listed and accepted by the process owner
- ☐ The parity test plan is written

Stage 5 · Build

- ☐ The flow is built in a solution in Development, with environment variables and connection references
- ☐ The child flows are built first; every action is named after its Nintex ID
- ☐ A second developer has reviewed the build

Stage 6 · Prove parity

- ☐ Every scenario has been run in both systems
- ☐ The results are compared and the differences explained
- ☐ The process owner has signed off

Stage 7 · Go live

- ☐ The managed solution is deployed to Production
- ☐ Users are told what changes for them
- ☐ New starts go to Power Automate, and the Nintex start option is removed
- ☐ The first runs are checked

Stage 8 · Operate and improve

- ☐ Monitoring and support are in place
- ☐ The defect fixes are planned
- ☐ When the old runs have finished: Nintex history archived, workflow removed, inventory updated

13. Glossary

Term	Meaning
Nintex Workflow 2016	The add-on that runs most of our current workflows inside SharePoint 2016
Power Automate	Microsoft's cloud service for automated flows; the replacement for SharePoint workflows
Flow	A Power Automate process, the equivalent of a workflow
Child flow	A flow that another flow calls, used for reusable steps
User-defined action (UDA)	A reusable Nintex building block; becomes a child flow or a copied scope
Solution	A package that holds flows and their settings, and moves them between environments
Environment	A separate Power Platform space: Development, Test or Production
Environment variable	A setting, such as a site address, that differs per environment
Connection reference	The account a flow uses to connect to SharePoint, Outlook or Teams
Parity	Behaving exactly like the old workflow
Parity note	A mark in the rebuild guide where a known defect is copied on purpose
Deviation	A difference from Nintex that Power Automate cannot avoid
On-premises data gateway	Software that lets cloud flows reach SharePoint 2016 in our own data centre
Wave	A group of workflows moved together

14. Sources

- Microsoft: SharePoint Migration Tool – workflow migration overview. The tool converts SharePoint Designer and out-of-the-box workflows only, not Nintex, and the result can behave differently from the original.
- Microsoft 365 Message Center MC542767 – SharePoint 2013 workflow retirement (2 April 2026).
- Microsoft 365 Message Center MC616550 – InfoPath retirement (14 July 2026).
- Microsoft: Create child flows and the asynchronous flow pattern (the 120-second response limit).
- Nintex Community: End-of-life schedule for Nintex Workflow for SharePoint Server 2016.
- In this repository: workflow analysis, rebuild guide, InfoPath form analysis.